OPTIMAL WINDOWED TEST FUNCTIONS FOR BOUNDING ANALYTIC RANKS OF ELLIPTIC CURVES

SABRINA HACKFORT (MEG)

Abstract. Assuming the Generalized Riemann Hypothesis, the explicit formula for the L-function of an elliptic curve $E/\mathbb{Q}$ bounds the analytic rank of E in terms of the traces a_p for $p < e^{2L}$, for any admissible test function supported on a window $[-L, L]$. In rank computations this test function is usually fixed in advance, typically to a Fejér-type kernel; optimising it for an individual L-function goes back to Booker, and an earlier test on quadratic twists of rank about 7 found the gain to be small. We choose, for each individual curve, the test function that minimises the resulting bound within a finite-dimensional space supported on $[-L, L]$, certify the result, and measure the gain systematically, up to the current record curves. The minimiser is the solution of a linear system with the Gram matrix of the Weil quadratic form of E, and the optimal bound equals the reciprocal of the corresponding reproducing kernel at the central point, which we call the *radix Riemanniana* of E. Because the Fejér-type kernel lies in this space, the optimised bound is never weaker. We certify every bound with ball arithmetic, check the prime-side evaluation against the zero side computed independently from the zeros of $L(E, s)$, and benchmark against the fixed kernel on 1642 curves from Cremona's tables, 2500 curves from the Stein–Watkins database and a set of curves with conductors up to $3 \cdot 10^{16}$. Depending on the sample and on the stopping criterion, the number of primes required to pin down the analytic rank decreases by 8–23% in aggregate. For individual curves of large conductor the decrease ranges from 0% to 43%. Within each sample the gain is largest at rank 0 and decreases with the rank; it grows again, and by a much larger factor, in the regime of large rank and large conductor where the computation is most expensive. On the smallest-conductor curves of rank 28–31 in the ICARM leaderboard the optimised kernel needs 84–85% fewer primes to certify the rank. For the current record, the rank-32 curve #4239 of Elkies, Klagsbrun and Li, it certifies $r_{\mathrm{an}} \leq 32$ under GRH with $2.53 \cdot 10^9$ primes — the first upper bound for this curve — and hence $r_{\mathrm{an}} = 32$ under GRH and BSD, while the fixed kernel, certified at $L = 12.40$ with 2 481 819 193 primes, is still 0.39 above the target. On all 395 leaderboard curves of rank 24–31, with primes below 10^{10}, the optimised kernel certifies $r_{\mathrm{an}} \leq r$ on 379 curves and the fixed kernel on 372; where both succeed the fixed kernel needs a median of 4.0 times as many primes. We validate the fixed-kernel baseline by reproducing every published value of Bober and of Klagsbrun, Sherman and Weigandt with an independent implementation.

1. Introduction

Let $E/\mathbb{Q}$ be an elliptic curve of conductor N and let $r_{\mathrm{an}}(E)$ be the order of vanishing of $L(E, s)$ at the central point. Under the Generalized Riemann Hypothesis (GRH), Mestre's explicit-formula method [13] bounds $r_{\mathrm{an}}(E)$ from above by a quantity computed from the traces of Frobenius a_p for p below a cutoff. Bober [4] made this practical with a Fejér-type test function, and Klagsbrun, Sherman and Weigandt [10] used the same function to bound the analytic rank of the Elkies curve by 28 under GRH. Their accompanying result that its Mordell–Weil rank equals 28 rests on computing the 2-ranks of class groups of associated cubic

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ORCID: 0009-0007-7458-2487.

fields, not on the explicit formula; the method discussed here bounds the analytic rank only. In these works the shape of the test function is fixed, and only its support (the cutoff) is varied.

This note asks a simple question: *for a given curve and a given cutoff, which test function gives the best bound?* Within any finite-dimensional space of admissible functions supported on $[-L, L]$ the answer is explicit. The explicit formula expresses the bound as a ratio of a quadratic form (the Weil form of E, evaluated through the a_p) to the square of a linear functional. The minimiser is therefore obtained from one linear solve with a Gram matrix. The optimal value is the reciprocal of the reproducing kernel at 0 of the space equipped with the Weil inner product.

What is known, and what is new. The Weil quadratic form restricted to functions supported on a finite interval, and its Gram matrices with respect to a basis, have been studied for the Riemann zeta function by Yoshida [19], by Connes, Consani and Moscovici [6], by Suzuki [17] via screw functions and de Branges spaces, and recently by Shi [15] as finite "Prime–Weil matrices". Reproducing-kernel methods for Fourier optimisation problems attached to *families* of L-functions, where the input is the limiting one-level density rather than data of an individual L-function, are due to Carneiro, Chirre and Milinovich [5].

Optimising the test function for an *individual* L-function is not new. Booker [3, §3] writes the explicit-formula sum over zeros as a positive definite quadratic form in the coefficients of a finite family of test functions and minimises it under a linear normalisation by inverting the matrix, in order to locate zeros of Artin L-functions. Watkins, Donnelly, Elkies, Fisher, Granville and Rogers [18, §7] applied this extension to analytic-rank bounds for quadratic twists and found the improvement to be small, typically about 0.05 (for example 7.138 to 7.090 with 10 basis functions, 7.089 with 40), and therefore used the fixed kernel. Proposition 3.1 below is the same minimisation in a Legendre basis on the window. We thank the literature for this and claim no novelty for the principle.

What this note adds is (i) certified output for every bound, (ii) a systematic measurement of the gain against the fixed kernel on several thousand curves, and (iii) the observation that the gain, small in the range tested in [18], becomes large exactly on curves of high rank and large conductor, where these computations are most expensive. On the ICARM high-rank leaderboard this turns into certified upper bounds for curves where none were known, including the current rank-32 record (Section 8).

Results in brief.

- The optimised bound is never weaker than the fixed-kernel bound at the same cutoff (Proposition 3.4). This is a consequence of the construction, not an empirical finding. The empirical question is *how much* it gains.
- Every reported bound is certified rigorously: in Arb ball arithmetic, or, on the leaderboard sweep, with the prime sum in outward-rounded interval arithmetic and the remaining terms in Arb. In the benchmark files of Sections 6 and 7 the two endpoints of each enclosure coincide when printed as doubles, because at those prime counts the radius falls below double precision; the certified decision is the comparison of the upper endpoint with the target. No certified bound in our tests contradicts the rank recorded for the curve.
- The prime-side value of the Weil form agrees with the zero-side value computed from the zeros of $L(E, s)$. The discrepancy halves with each doubling of the height up to which zeros are included (Section 5).
- The number of primes needed to determine the analytic rank (Section 6) decreases in aggregate by 8.3–15.2% (Cremona, $N < 4.1 \cdot 10^5$) and by 17.0–22.7% (Stein–Watkins,

$4.4 \cdot 10^6 \le N \le 7.5 \cdot 10^9$), depending on the stopping criterion. For individual curves of conductor up to $5 \cdot 10^{15}$ the decrease ranges from 0% to 43%.

- On curves of record rank (Section 8) the gain is much larger. For the smallest-conductor curves of ranks 28–31 in the ICARM leaderboard, the number of primes needed for a bound below $r+2$ drops by a factor of about 6.5. For the bound below $r+1$ the fixed kernel fails for three of the four curves with all primes below 10^9, while the optimised kernel succeeds with between $2.3 \cdot 10^5$ and $1.3 \cdot 10^7$ primes. On the historic record curves E_{20}–E_{27} the factor is 2–3.5.
- For the current record curve #4239 (rank ≥ 32) the optimised kernel certifies $r_{\mathrm{an}} \le 32$ under GRH with all 2 529 822 599 primes below $6.02 \cdot 10^{10}$ — and $r_{\mathrm{an}} = 32$ under GRH and BSD, using the 32 independent points of the leaderboard. The fixed kernel does not reach the target: its certified bound at $L = 12.40$, with 2 481 819 193 primes, is $[34.39360281 \pm 4.6 \cdot 10^{-9}]$ (Section 8).
- On the 395 leaderboard curves of rank 24–31 ($p < 10^{10}$) the optimised kernel certifies $r_{\mathrm{an}} \le r$ on 379 curves, the fixed kernel on 372; 7 only by the optimised kernel, none only by the fixed one (Section 9).
- The fixed-kernel baseline is exactly Bober's method: an independent implementation of Bober's formula reproduces all seven published values in [4, 10]; it agrees with our floating-point evaluation to 10^{-6}, the precision to which we tabulate it, and lies inside our Arb enclosures, whose radii are below 10^{-12} (Section 7).

2. The explicit formula and rank bounds

Write $L(E,s) = \sum_{n\ge1} a_n n^{-s-1/2}$ so that the central point is $s = 1/2$, and put

$$\Lambda(E,s) = N^{s/2}(2\pi)^{-s}\,\Gamma(s+\tfrac12)\,L(E,s), \qquad \Lambda(E,s) = \pm\Lambda(E,1-s).$$

For $p \nmid N$ let α_p, β_p be defined by $\alpha_p + \beta_p = a_p/\sqrt{p}$, $\alpha_p\beta_p = 1$, and set $c_{p^k} = \alpha_p^k + \beta_p^k$. For $p \mid N$ set $c_{p^k} = (a_p/\sqrt{p})^k$ (so $c_{p^k} = 0$ for additive reduction). Then $-\frac{L'}{L}(E,s) = \sum_{n\ge2} \Lambda(n) c_n n^{-s}$ in the normalisation above.

Let $f\colon \mathbb{R} \to \mathbb{R}$ be even, bounded and supported in $[-L, L]$. Put $g = f * f$, so that g is even and supported in $[-2L, 2L]$, and

$$\hat f(r) = \int_{\mathbb{R}} f(x) e^{irx}\,dx, \qquad h(r) = \hat g(r) = \hat f(r)^2.$$

Writing the nontrivial zeros as $\rho = \frac12 + i\gamma$ (with multiplicity), the explicit formula reads

$$\sum_{\gamma} h(\gamma) \;=\; g(0)\log\frac{N}{4\pi^2} \;+\; 2\Big[-\gamma_{\mathrm{E}}\,g(0) + \int_0^\infty \frac{g(0)-g(u)}{e^u-1}\,du\Big] \;-\; 2\sum_{n\ge2}\frac{\Lambda(n)\,c_n}{\sqrt{n}}\,g(\log n), \tag{1}$$

where γ_{E} is Euler's constant. Only prime powers $n < e^{2L}$ contribute. We denote the right-hand side by $W_E(f)$.

Lemma 2.1. *Assume GRH for $L(E,s)$. Then for every nonzero f as above,*

$$r_{\mathrm{an}}(E) \;\le\; \frac{W_E(f)}{\hat f(0)^2}.$$

Proof. Under GRH every γ is real. Since f is real and even, $\hat f$ is real, so $h(\gamma) = \hat f(\gamma)^2 \ge 0$. The central zero contributes $r_{\mathrm{an}}(E)\,\hat f(0)^2$ to the left-hand side of (1), and every other term is nonnegative. □

Remark 2.2. For $f = \mathbf{1}_{[-L,L]}$ one has $g(u) = (2L - |u|)_+$ and $h(r) = 4\sin^2(Lr)/r^2$. Up to scaling, this is the Fejér-type test function used in [4, 10]. We call the resulting bound $B_{\mathrm{std}}(L)$. With $L = \pi\Delta$ it coincides with Bober's $\sum_\gamma f(\gamma;\Delta)$: $\hat f(r)^2/\hat f(0)^2 = (\sin(\Delta\pi r)/(\Delta\pi r))^2$, the conductor term becomes $\log(\sqrt{N}/2\pi)/(\pi\Delta)$, and the archimedean term evaluates to Spicer's closed form $-\gamma_{\mathrm{E}}/(\pi\Delta) + \big(\pi^2/6 - \mathrm{Li}_2(e^{-2\pi\Delta})\big)/(2\pi^2\Delta^2)$ quoted in [10].

3. The optimal window kernel

The map $f \mapsto W_E(f)$ is a quadratic form, $W_E(f) = \mathcal{B}_E(f,f)$, where $\mathcal{B}_E$ is obtained from (1) by replacing $g = f * f$ with the symmetrised correlation of two functions. Fix the basis

$$b_k(x) = P_{2k}(x/L)\,\mathbf{1}_{[-L,L]}(x), \qquad 0 \le k < K,$$

of even polynomials on the window, where P_m is the Legendre polynomial of degree m. Let $G_{ij} = \mathcal{B}_E(b_i, b_j)$ be the Gram matrix and $v_i = \hat b_i(0) = \int b_i$, so that $v = (2L, 0, \dots, 0)$. For $f = \sum c_k b_k$ we have $W_E(f) = c^{\mathsf T} G c$ and $\hat f(0) = v^{\mathsf T} c$.

Proposition 3.1. *If G is positive definite, the minimum of $W_E(f)/\hat f(0)^2$ over nonzero f in* $\mathrm{span}(b_0, \dots, b_{K-1})$ *equals*

$$B_{\mathrm{rad}}(L) = \frac{1}{v^{\mathsf T} G^{-1} v},$$

and is attained at $c^* = G^{-1}v/(v^{\mathsf T}G^{-1}v)$.

Proof. Minimise $c^{\mathsf T}Gc$ subject to $v^{\mathsf T}c = 1$ (Lagrange multipliers, or Cauchy–Schwarz in the inner product $\langle x, y\rangle = x^{\mathsf T}Gy$). □

Since $v = \Phi(0)$, the quantity $v^{\mathsf T}G^{-1}v$ is the value at the central point of the following kernel.

Definition 3.2 (Radix Riemanniana)**.** Let $\Phi(z) = (\hat b_0(z), \dots, \hat b_{K-1}(z))^{\mathsf T}$. For an elliptic curve $E/\mathbb{Q}$ and a window $[-L, L]$, the *radix Riemanniana* of E (of order K) is the kernel

$$\mathcal{R}_{E,L}(w, z) \;=\; \Phi(w)^*\, G^{-1}\, \Phi(z),$$

that is, the reproducing kernel of the space $\{\hat f : f \in \mathrm{span}(b_k)\}$ equipped with the Weil inner product $\mathcal{B}_E$. It is built from the Gamma factor, the conductor and the primes $p < e^{2L}$ alone, without reference to the zeros of $L(E, s)$. The optimal bound of Proposition 3.1 is

$$B_{\mathrm{rad}}(L) = \frac{1}{\mathcal{R}_{E,L}(0,0)}.$$

Remark 3.3. As $K \to \infty$ the radix Riemanniana is the window analogue, for $L(E, s)$, of the reproducing kernels of the Hilbert spaces of entire functions attached to the Weil form of ζ in [17]. The name and the construction from prime data go back to the author's work on Weil positivity for ζ.

Proposition 3.4. *For every L with G positive definite, $B_{\mathrm{rad}}(L) \le B_{\mathrm{std}}(L)$.*

Proof. $\mathbf{1}_{[-L,L]} = b_0$ lies in the span, so $B_{\mathrm{std}}(L)$ is one of the values minimised over in Proposition 3.1. □

Positive definiteness of G holds under GRH: $W_E(f) = \sum_\gamma \hat f(\gamma)^2 \ge 0$, and it vanishes only if $\hat f$ vanishes on the whole zero set; since $\hat f$ is entire of exponential type $2L$ while the zeros of $L(E, s)$ have counting function $\sim (T \log T)/\pi$, this forces $f = 0$. In practice we simply attempt a Cholesky factorisation and discard the window if it fails. Note that *the certificate in Section 4 does not depend on optimality or on positive definiteness*: it evaluates $W_E(f)$ for the specific f that was found.

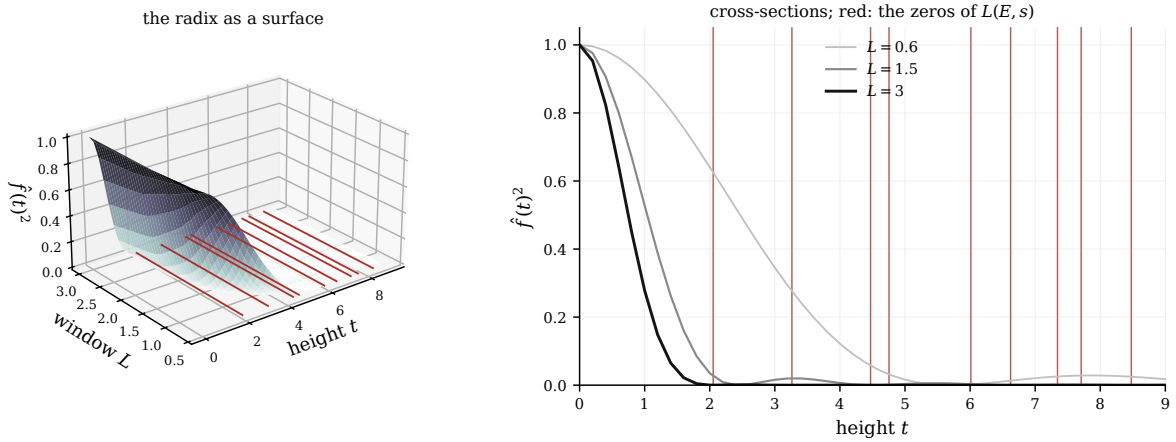


FIGURE 1. The radix Riemanniana of $5077a1$ (rank 3, $N = 5077$): the optimal kernel $\hat{f}(t)^2$, normalised to $\hat{f}(0) = 1$, over the height t and the window L, with the zeros of $L(E, s)$ in red. As the window widens the kernel concentrates at the central point and pulls away from the zeros; the bound $1/\mathcal{R}_{E,L}(0,0)$ falls from 5.4241 at $L = 0.6$ to 3.0000 at $L = 3$, the rank of the curve. Computed from the same Gram matrices as the bounds, by polarisation of the form implemented in `scripts/certify.py`.

4. CERTIFICATION

The optimisation is carried out in double precision. The resulting coefficients are converted to exact rationals $c_k \in \mathbb{Q}$, and L is taken to be an exact decimal. For this fixed $f = \sum c_k b_k$ we evaluate $W_E(f)/\hat{f}(0)^2$ rigorously in ball arithmetic (Arb [1] via `python-flint`, 256-bit precision):

(1) On $[0, 2L]$, $g = f * f$ is a polynomial with rational coefficients. We compute it exactly in $\mathbb{Q}[u]$ and obtain $\hat{f}(0)$ exactly.
(2) The archimedean integral is split at $u = 1/4$. On $[0, 1/4]$ we integrate $(g(0) - g(u))/u$ against the Bernoulli expansion $u/(e^u - 1) = \sum_n B_n u^n/n!$, truncated after 80 terms. The tail is bounded with $|B_n|/n! \le 4(2\pi)^{-n}$. On $[1/4, 2L]$ the integrand is analytic and is integrated rigorously by Arb's certified Gauss–Legendre integration. The constants γ_{E}, π and the logarithms are balls.
(3) The prime sum runs over all prime powers $n < e^{2L}$. The quantities $\log n$, $\sqrt{n}$ and c_n are balls computed from the exact integers a_p. If the ball for $\log n$ contains $2L$, the window is rejected.

The output is a ball containing $W_E(f)/\hat{f}(0)^2$. A statement $r_{\mathrm{an}}(E) < r + 1$ is recorded only if the upper end of the ball is below $r + 1$.

Large prime counts. For the record curves of Section 8 the prime sum has up to $2.5 \cdot 10^9$ terms, and two further tools are used.

Traces of Frobenius. The a_p for $p < 10^9$ (Sections 6–7) are computed with PARI/GP [14]. Beyond that we use our own implementation of the baby-step giant-step method of Shanks and Mestre on the curve and its quadratic twist: the whole Hasse interval is searched, the parity of a_p is read off from the 2-torsion of $E(\mathbb{F}_p)$, and candidate sets from several points are intersected until a single value remains; the program never guesses and aborts otherwise. Every

value is finally re-checked on fresh points of E and of its twist before it is written. On #4239 it reproduces all 50 847 534 values of PARI for $p < 10^9$ bit for bit.

Prime sum. The sum in step 3 is enclosed by a second, independent implementation in `f64` interval arithmetic: every operation is rounded to nearest and widened outward by one unit in the last place, $\log p$ is computed from the exact reduction $p = m2^e$ and a bounded series for artanh, $\log 2$ is an interval checked against Arb, and g enters through its exact Chebyshev coefficients on $[0, 2L]$, rounded outward. All other parts of $W_E(f)/\hat{f}(0)^2$ are still computed in Arb. On every curve on which both were run the two certificates have the same midpoint and the same verdict; the interval version is about 10 times faster per core and its enclosures are wider, far below every margin reported here.

Which of the two is used is decided by the script, not by the prime bound. Tables 6, 7 and 8, that is the record and historic curves including #4239 at $p < 6.02 \cdot 10^{10}$, are certified in Arb alone (`highrank/cert_4239.py`, `cert_stream.py`). The board of Section 9 is certified throughout with the interval version (`scripts/cert_fast.py`), at every prime bound. Its radii are flat at a median of $2.6 \cdot 10^{-9}$ for $p < 10^6$ through $p < 10^9$ — there the working precision of the Arb part dominates — and rise to a median of $6.5 \cdot 10^{-8}$ (largest $4.9 \cdot 10^{-7}$) at $p < 10^{10}$, where the $4.5 \cdot 10^8$ outward-rounded terms of the prime sum take over. On every one of the 790 certified board entries the enclosure is separated from the target by at least 440 times its own radius (the tightest case is the fixed kernel on #3871 at $p < 10^{10}$: margin $1.45 \cdot 10^{-4}$, radius $3.29 \cdot 10^{-7}$, ratio 440.7), so the width of the enclosure decides no case.

5. Validation against the zeros

Identity (1) lets us check the whole pipeline (Rust optimiser → rational coefficients → certified prime side) against an independent computation. We compute the zeros γ of $L(E, s)$ up to height T with PARI's `lfunzeros` and evaluate the zero side $r_{\mathrm{an}}\hat{f}(0)^2 + \sum_{0<|\gamma|\le T} \hat{f}(\gamma)^2$ for the same certified f. Since every omitted term is nonnegative, the truncated zero side must lie below the prime side, and the difference must tend to 0 as $T \to \infty$.

curve	r_{an}	L	prime side (certified)	zero side, $\lvert\gamma\rvert \le T$	T	difference
37a1	1	0.40	1.9967148280	1.9923507969	600	$4.4 \cdot 10^{-3}$
389a1	2	0.65	2.9926821162	2.9903944027	600	$2.3 \cdot 10^{-3}$
5077a1	3	0.90	3.9514652620	3.9497072474	600	$1.8 \cdot 10^{-3}$
234446a1	4	1.15	4.9323755046	4.9292024883	300	$3.2 \cdot 10^{-3}$

Table 1. Certified prime side versus zero side, both normalised by $\hat{f}(0)^2$, for the optimised test function at the cutoff where the bound first drops below $r_{\mathrm{an}} + 1$.

For 37a1 the difference is $8.1 \cdot 10^{-3}$, $4.4 \cdot 10^{-3}$ and $2.4 \cdot 10^{-3}$ at $T = 300$, 600 and 1200 (446, 1024 and 2313 zeros). It roughly halves with each doubling of T, consistent with a truncation tail of order $(\log T)/T$.

6. Benchmarks

For each curve we scan L on a grid of step 0.05 with $K = 10$ basis functions. For both B_{std} and B_{rad} we record the first L at which the certified bound drops below $r_{\mathrm{an}} + 1$ (criterion A), and the first L at which it drops below $r_{\mathrm{an}} + 2$ (criterion B; together with the root number, this already determines the parity of r_{an}). Cost is measured as the number of primes $p < e^{2L}$ used.

Because of the grid, prime counts at the threshold are quantised. Ties are therefore frequent and slightly understate the gain.

6.1. **Cremona's tables.** We take one curve per isogeny class, sampled up to 150 per rank from three blocks of Cremona's tables [7] ($61 \le N \le 409989$, 1642 curves, ranks 0–3), We provide a_p for $p < 8200$ (1028 primes) and cut off at $L \le 4.5$. Table 2 reports the results. In total 6449 bounds were certified, and none contradicts the rank recorded for the curve.

	curves decided (std / rad)	rad fewer / equal / more	primes saved
criterion A ($< r_{\mathrm{an}} + 1$)	1573 / 1592	559 / 1014 / 0	15.2%
criterion B ($< r_{\mathrm{an}} + 2$)	1642 / 1642	263 / 1379 / 0	8.3%

criterion A, by rank	0	1	2	3
curves (both decided)	381	450	450	292
primes saved	18.8%	4.3%	3.9%	4.5%

TABLE 2. Cremona sample. "Primes saved" is the pooled ratio $1 - \sum_E P_{\mathrm{rad}}(E) / \sum_E P_{\mathrm{std}}(E)$ over the curves decided by *both* kernels, that is the total work saved on the sample; it is not the saving on a typical curve. The median per-curve saving is 0% under both criteria, and a saving occurs at all on 36% (A) and 16% (B) of the curves — see the text. "Fewer / equal / more" is likewise over the curves decided by both; "curves decided" counts each kernel separately. Under criterion A the optimised kernel decides 19 curves that the fixed kernel does not decide within the same prime bound, and there is no curve the other way round.

The two figures differ sharply, and the reason is worth stating. On a curve that both kernels decide with a few dozen primes the prime bound moves in steps of one prime, so the two usually tie; the median curve of the Cremona sample therefore saves nothing. The pooled figure is carried by the minority of curves on which the computation is actually expensive. That is the regime the method is meant for, and Sections 8 and 9 are about it; but the aggregate should not be read as a per-curve expectation.

6.2. **The Stein–Watkins database.** From the Stein–Watkins database [16] (files of the 2011 distribution) we sample 250 curves per (analytic) rank $0, 1, 2, 3, \ge 4$ from two groups: general conductors $4.4 \cdot 10^6 \le N \le 9.7 \cdot 10^7$ and prime conductors $3.6 \cdot 10^9 \le N \le 7.5 \cdot 10^9$, giving 2500 curves in total. We check every conductor and root number with PARI. The ranks in the database are Watkins' numerically determined analytic ranks. We provide a_p for $p < 10^5$ (9592 primes) and cut off at $L \le 5.75$. In total 9105 bounds were certified, and none contradicts the database rank.

6.3. **Large conductors and high rank.** Table 4 lists every curve of an additional test set, including the cases without gain. Prime counts up to e^{16} were handled by streaming the prime sum.

7. Validation against published fixed-kernel computations

To make sure that B_{std} is *exactly* the method of [4, 10], and not a variant of it, we re-implemented Bober's formula [4, eq. (3)] directly from the paper, with the archimedean term in Spicer's closed form and without any code shared with our optimiser. Conductors were taken

	decided (std / rad)	rad fewer / equal / more	primes saved
criterion A	2059 / 2096	1269 / 790 / 0	17.0%
criterion B	2466 / 2484	871 / 1595 / 0	22.7%

criterion B, by rank	0	1	2	3	≥ 4
curves (both decided)	466	500	500	500	500
primes saved	24.7%	15.1%	5.7%	3.1%	2.7%

TABLE 3. Stein–Watkins sample. "Primes saved" is the pooled ratio as in Table 2. The median per-curve saving is 7.0% under criterion A and 0% under criterion B; a saving occurs at all on 62% and 35% of the curves.

curve	N	rank	criterion	primes std / rad	saved
twist of 11a1, $d = 10009$	1101980891	1	B	329 / 302	8%
twist of 389a1, $d = 10009$	38970051509	1	B	3523 / 3523	0%
twist of 11a1, $d = 100049$	110107826411	0	B	2466 / 2065	16%
twist of 5077a1, $d = 10009$	508614271237	1	B	1220 / 1019	16%
twist of 389a1, $d = 100049$	3893813133989	0*	B	17885 / 13613	24%
twist of 5077a1, $d = 100049$	50819766789877	1	B	1220 / 1220	0%
twist of 389a1, $d = 1000033$	389025674423621	0*	B	102084 / 77395	24%
twist of 5077a1, $d = 1000033$	5077335087528853	1	B	340436 / 195040	43%
twist of 11a1, $d = 1000033$	11000726011979	0	B	undecided by both for $L \leq 7.5$	
5187563742a1	5187563742	6	A	11 / 10	9%
$[1, -1, 0, -106384, 13075804]$	249649566346838	8	A	21 / 18	14%
$[1, -1, 0, -135004, 97151644]$	32107342006814614	9	A	32 / 29	9%

TABLE 4. Additional curves. Ranks are lower bounds from PARI's `ellrank` (points). *Here the 2-descent leaves rank 0 or 2 open. The root number $+1$ together with the certified bound $r_{\mathrm{an}} < 2$ gives $r_{\mathrm{an}} = 0$ under GRH, and hence rank 0 by Kolyvagin [12].

from the ICARM leaderboard [9]; $\log N$ agrees with [4, Table 1] to all printed digits. Table 5 shows that the independent implementation reproduces every published value we could find. The floating-point $B_{\mathrm{std}}(\pi\Delta)$ of our optimiser agrees with it to 10^{-6} on all seven curves, which is the precision to which we tabulate the independent values. On the three curves certified in Arb the independent value lies inside the enclosure, which has radius below 10^{-12}; the difference between the two midpoints is $2.7 \cdot 10^{-8}$, $1.6 \cdot 10^{-7}$ and $2.8 \cdot 10^{-7}$ and reflects the rounding of the tabulated value, not the certificate.

8. Curves of record rank

For curves of large rank the cutoff, and with it the number of primes, grows quickly: Klagsbrun, Sherman and Weigandt used $\Delta = 4$ (primes up to $e^{8\pi} \approx 8 \cdot 10^{10}$, about 40 hours) for the Elkies curve [10], and the leaderboard record of rank 30 was certified with $\Delta = 4.25$ (primes up to $\approx 4 \cdot 10^{11}$) [9]. This is where a better test function matters most. The following comparison was suggested by Z. Klagsbrun.

Setup. All curves are taken from the ICARM Elliptic Curve Rank Leaderboard [9] (snapshot of 12 October 2026, 4201 curves) in their global minimal models; the rank r is the leaderboard's

curve	$\log N$	Δ	published	independent	certified $B_{\mathrm{std}}(\pi\Delta)$
E_{20}	170.09	2.0	21.70 [4]	21.690724	$[21.6907238431 \pm 5 \cdot 10^{-13}]$
E_{21}	196.68	2.5	22.68 [4]	22.672713	
E_{22}	182.72	2.0	23.71 [4]	23.704751	$[23.7047512825 \pm 3 \cdot 10^{-13}]$
E_{23}	205.06	2.5	24.49 [4]	24.483411	
E_{24}	219.93	2.5	25.57 [4]	25.568173	$[25.5681730272 \pm 5 \cdot 10^{-13}]$
E_{27}	287.01	3.1	< 29 [10]	28.988058	
E_{28}	325.90	3.2	31.30 [4]	31.298359	

TABLE 5. Reproduction of published values of $\sum_\gamma f(\gamma;\Delta)$; [4] rounds up. E_{20}–E_{24}, E_{28} are the curves of [4], E_{27} that of [10].

certified lower bound. We recomputed every conductor from local data with PARI and checked that the root number equals $(-1)^r$ in every case, so that a certified bound $B < r+2$ gives $r_{\mathrm{an}} \le r$ under GRH (and rank exactly r under GRH and BSD). We computed a_p for all $p < 10^9$ (50 847 534 primes), scanned L with step 0.01 and $K = 10$ (for E_{28}, $K = 14$ and $K = 20$ give the same values to 10^{-4}), and certified every reported value with Arb as in Section 4.

Ranks 28–31. For each rank 28–31 we take the curve of smallest conductor. Table 6 lists the number of primes needed. For the target $r+2$ the optimised kernel reaches it in all four cases with a window shorter by $\Delta L \approx 1.0$, which reduces the prime bound by roughly e^2. Near $r+1$ the fixed-kernel bound flattens, and the gap becomes much larger.

			bound $< r+2$		bound $< r+1$	
ICARM	r	$\log N$	fixed	optimal	fixed	optimal
#3762	28	242.9	122 776	**19 235**	21 600 954	**234 855**
#3835	29	265.4	316 068	**48 890**	$> 5.08 \cdot 10^7$ [a]	**717 477**
#3897	30	279.9	387 811	**59 820**	$> 5.08 \cdot 10^7$ [a]	**691 123**
#3900	31	325.3	4 340 969	**665 760**	$> 5.08 \cdot 10^7$ [a]	**13 209 786**

TABLE 6. Number of primes $p < e^{2L}$ at the first certified L (grid step 0.01). [a]Not reached with all primes $p < 10^9$; the certified fixed-kernel bounds at $L = 10.36$ are, in the order of the rows, $[30.00290409 \pm 3.2 \cdot 10^{-9}]$ for #3835, $[31.08984065 \pm 3.1 \cdot 10^{-9}]$ for #3897 and $[32.43064260 \pm 2.5 \cdot 10^{-10}]$ for #3900.

The historic record curves. On the record curves of [4, 10] the gain is smaller but consistent (Table 7). For E_{27} the minimal fixed-kernel window $L = 9.72$ ($\Delta = 3.09$) matches the choice $\Delta = 3.1$ of [10]. For E_{28} neither kernel reaches a bound below 30 with $p < 10^9$ (at $L = 10.36$: 31.092 fixed, 30.820 optimal).

The 2026 records. In August 2026 curves of rank at least 30 (#273) and 31 (#302) were found by L. Alpöge and A. Howell [9]. For #273 ($\log N = 339.35$, root number $+1$) the leaderboard reports a fixed-kernel bound below 32 with $\Delta = 4.25$. The optimised kernel certifies

$$r_{\mathrm{an}}(\#273) \;\le\; [31.99478129 \pm 4.5 \cdot 10^{-9}] \;<\; 32$$

at $L = 9.66$, with 13 461 681 primes $p < 2.46 \cdot 10^8$, so $r_{\mathrm{an}}(\#273) \le 30$ under GRH and $r_{\mathrm{an}} = 30$ under GRH and BSD. The fixed kernel does not reach 32 with all primes below 10^9: its certified value at $L = 10.36$ is $[32.08095018 \pm 1.5 \cdot 10^{-9}]$; carried to the cap of the board of Section 9 it

	E_{20}	E_{21}	E_{22}	E_{23}	E_{24}	E_{27}
fixed, $< r+2$	11 988	181 084	11 988	77 395	103 973	15 078 540
optimal, $< r+2$	**5 515**	**83 298**	**5 133**	**23 068**	**32 069**	**4 340 969**
optimal, $< r+1$	47 114	—	37 128	353 398	433 671	—

TABLE 7. Historic record curves, primes needed (grid step 0.01, all values certified). The $r+1$ row was searched only up to $p < 6.7 \cdot 10^6$ (457 497 primes); within that range the fixed kernel reaches no bound below $r+1$ on any of the six curves, and "—" marks the two on which the optimised kernel does not either. The $r+2$ row was searched over a larger range, up to $1.5 \cdot 10^7$ primes.

succeeds at 72 713 579 primes, so the optimised kernel needs 18.5% of them. The leaderboard entry used the fixed kernel at $\Delta = 4.25$ without scanning L, that is about $1.5 \cdot 10^{10}$ primes; most of the factor 1100 between that and our $1.3 \cdot 10^7$ is the scan, not the optimisation. For #302 ($\log N = 375.22$, root number -1) neither kernel reached 33 with $p < 10^9$ (at $L = 10.36$: fixed $[34.02864458 \pm 1.3 \cdot 10^{-9}]$, optimal 33.566), and a fit of the form $A + C/L$ to the optimal bound suggested $p \lesssim 6 \cdot 10^9$. Extending the computation to $p < 10^{10}$ confirms this: the optimised kernel certifies

$$B_{\mathrm{rad}}(\#302) \;=\; [32.9991320 \pm 7.09 \cdot 10^{-8}] \;<\; 33,$$

so $r_{\mathrm{an}}(\#302) \le 31$ under GRH, and $r_{\mathrm{an}} = 31$ under GRH and BSD. At the cap the fixed kernel, using 452 518 634 primes against 339 885 370 for the optimised one, is still at $[33.349252 \pm 4.92 \cdot 10^{-7}]$; the fit had suggested $p \lesssim 3 \cdot 10^{10}$ for it, which remains an extrapolation.

The rank-32 record. On 15 October 2026 Elkies, Klagsbrun and Li added the curve #4239 of rank at least 32 to the leaderboard [9], a specialisation of a rank-17 fibration of the K3 surface of discriminant 1152, with $\log N = 414.086$ and root number $+1$; no upper bound was known. With all primes $p < 6.02 \cdot 10^{10}$ (a_p by the method above, Arb certification):

kernel	L	primes $p < e^{2L}$	certified bound
optimal	12.41	2 529 822 599	$[33.99820448 \pm 1.8 \cdot 10^{-9}]$
optimal	12.40	2 481 819 193	$[34.00293607 \pm 3.0 \cdot 10^{-9}]$
fixed	12.40	2 481 819 193	$[34.39360281 \pm 4.6 \cdot 10^{-9}]$

TABLE 8. The record curve #4239. The first row is the certificate.

Hence $r_{\mathrm{an}}(\#4239) \le 33$, and by parity $r_{\mathrm{an}}(\#4239) \le 32$ under GRH; with the 32 independent points of the leaderboard the rank is exactly 32 under GRH and BSD. The fixed kernel does not reach the target anywhere in the range computed: its certified bound at $L = 12.40$ is 0.39 above it (Table 8), and at its largest window $L = 12.44$, with 2 679 479 553 primes, its floating-point value is still 34.375. A linear extrapolation in L suggests it would need primes up to about $3 \cdot 10^{11}$. Doubling the basis from $K = 10$ to $K = 20$ changes the optimal bound at $p < 10^9$ only in the fifth decimal ($35.30281 \to 35.30276$): within the window the optimisation is saturated, and further gains have to come from more primes.

Note added (25 October 2026). After the bound above was posted to the leaderboard on 17 October 2026, R. Jain independently obtained $\operatorname{rank} E(\mathbb{Q}) \le 32$ for #4239 on 19 October 2026 by a different method [11]: a bound on the 2-Selmer group through the class groups of the cubic field of the 2-division polynomial, following [2, 10]. His bound concerns the Mordell–Weil

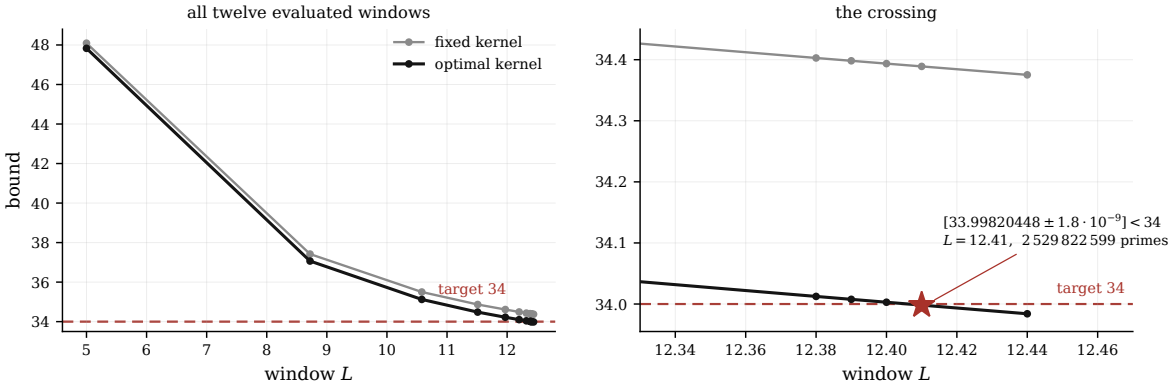


FIGURE 2. The two kernels on #4239 over the twelve windows at which both were evaluated (floating point; the entries of Table 8 are the certified values at $L = 12.40$ and 12.41). The optimised bound crosses the target between $L = 12.40$ and $L = 12.41$; the fixed-kernel bound is still above 34.37 at the largest window computed.

rank directly; it is conditional on GRH for finitely many Dedekind zeta functions and does not use BSD. The bound of this section concerns the analytic rank and is conditional on GRH for $L(E, s)$. The two computations share no code and rest on different hypotheses, so each confirms the value 32 independently of the other.

9. The ICARM high-rank board

Following a suggestion of Z. Klagsbrun, we ran both kernels on every curve of the leaderboard with rank lower bound 24–31 (snapshot of 17 October 2026). For each curve the prime bound is raised by factors of 10 from 10^6 until both kernels reach the target, or up to a cap; the target is $r + 2$; the root number equals $(-1)^r$ for all 395 curves, so a certified bound below $r + 2$ gives $r_{\mathrm{an}} \le r$ under GRH, and $r_{\mathrm{an}} = r$ under GRH and BSD. Every hit is certified as in Section 4; when a kernel does not reach the target below the cap, its certified bound at the largest window is recorded instead. Table 9 summarises the result by rank; the full list is in the repository (`highrank/board/`).

The median ratio grows with the rank, though not monotonically, from 3.1 at rank 24 to 6.9 at rank 31 (Table 9). On 16 curves neither kernel reaches the target with $p < 10^{10}$; on all of them the certified bound of the optimised kernel is the smaller one; their certified bounds at the cap are listed in the repository.

10. From primes to time

Counting primes is the natural measure for this method, because the traces a_p dominate the cost on curves of large conductor. It is not, however, the same as wall-clock time, and the difference matters.

In the board computation of Section 9 the prime bound is raised in powers of ten and the traces are recomputed from scratch at each stage. On the machine used (one core of an Apple M-series laptop) the measured times to produce all a_p below P were 0.5, 1, 5, 57 and 699 seconds for $P = 10^6, \dots, 10^{10}$. The staged search therefore costs the sum of the stages up to the one that succeeds, not the last stage alone: 762.5 seconds for a curve carried to the cap,

rank	curves	optimal	fixed	only optimal	median ratio
24	55	55	55	0	3.1
25	65	65	65	0	3.6
26	43	43	43	0	3.6
27	44	43	43	0	4.1
28	87	76	74	2	4.2
29	50	50	47	3	4.7
30	34	34	33	1	5.3
31	17	13	12	1	6.9
all	395	379	372	7	4.0

TABLE 9. ICARM curves of rank 24–31: number of curves on which each kernel certifies $r_{\mathrm{an}} \le r$ under GRH with $p < 10^{10}$; "only optimal" = certified by the optimal kernel but not by the fixed one; ratio = primes needed by the fixed kernel divided by primes needed by the optimal kernel, over curves where both succeed.

against 699 for its final stage, a factor of 1.09. Aggregated over the board the factor is 1.11; for a curve whose last stage is 10^7 it is 1.5, because the stage times are not proportional to P.

On a curve run to the cap the traces accounted for 762.5 of 819 seconds of wall-clock, that is 93%. The remaining 7% covers the L-scan for *both* kernels and the Arb certification at every stage, and therefore bounds the extra cost of the optimised kernel on that curve. This is a single measurement, and it is the case most favourable to the traces; for a curve that succeeds at 10^6 the certification dominates instead.

Assigning to each kernel the stage at which it first reaches the target, and costing that stage by the five measured times, gives over the 372 curves on which both succeed a total trace cost of 2.49 hours for the optimised kernel against 5.53 hours for the fixed one, a factor of 2.22. These hours are extrapolated from five timings, not measured curve by curve. The factor is markedly less than the 4.0 median of the per-curve prime ratios, and the reason is the granularity of the search: a curve that needs twice as many primes usually falls in the same decade, so the saving is not realised. On the 203 curves where the two kernels fall in different decades the picture is:

stages	curves	optimised	fixed
$10^6 \to 10^7$	79	0.5 s	1 s
$10^7 \to 10^8$	68	1 s	5 s
$10^8 \to 10^9$	43	5 s	57 s
$10^9 \to 10^{10}$	12	57 s	699 s
$10^8 \to 10^{10}$	1	5 s	699 s

So even among these the gain in time ranges from a factor of 2 to a factor of 140, and on 79 of them it is below the median gain in primes. The single curve in the last row is #4032 of rank 30.

The honest summary is therefore: in the regime where this method is used at all, the saving in primes translates into a saving in time of the same order, but only once the prime counts differ by enough to cross a decade. A search on a finer grid of prime bounds would recover more of it.

11. Limitations

- All bounds assume GRH for $L(E, s)$ and concern the analytic rank. Statements about the algebraic rank additionally use points, the root number and, for $r_{\mathrm{an}} \leq 1$, the theorems of Gross–Zagier [8] and Kolyvagin [12].
- The basis size is fixed at $K = 10$ and the cutoff grid at step 0.05 (0.01 in Sections 7 and 8). Larger K can only improve B_{rad}, and a finer grid would turn some ties into gains. We have not optimised either choice.
- Certification uses Arb, and for the prime sum of large instances an independent `f64` interval implementation (Section 4); both are written by the author and cross-checked against each other, not formally verified.
- The gain depends on the curve: on record-rank curves it ranges from a factor of about 2 (E_{20}, E_{21}) to 6.4–6.5 (ICARM, ranks 28–31) for a bound below $r + 2$; on the board the ratio of prime counts ranges from 1.87 to 14.95 over the 372 curves on which both kernels succeed. For E_{28} we do not reach the target and report only an extrapolation; on the board (Section 9) such cases are listed with their certified bound at the cap.
- Throughout, cost is measured as the number of primes, that is the number of traces a_p required. Section 10 relates this to wall-clock time; the ratio in time is smaller than the ratio in primes. The evaluation cost of the kernels themselves is not separated out: per prime the optimised kernel additionally evaluates Chebyshev moments, and per window it assembles and factors the Gram matrix, whereas the fixed kernel only accumulates a sum. Both are contained in the non-a_p share bounded in Section 10.
- Ranks in Section 8 are certified lower bounds from the leaderboard. Our bounds concern the analytic rank; equality with the algebraic rank requires BSD.

Data and code availability

Code (Rust optimiser, certification scripts), the sampled curve lists and all certificates are available at `https://github.com/sectio-aurea-q/radix-rank` (archived on Zenodo, DOI: 10.5281/zenodo.23299461). The folders `validation/` and `highrank/` contain the independent implementation of Bober's formula and the scripts and certified values for Sections 7 and 8. Running `./reproduce.sh quick` re-derives and re-certifies the bounds for seven reference curves in about one minute; `scripts/rank_pipeline.sh` and `scripts/batch_icarm.py` reproduce Sections 8 and 9 from the leaderboard data.

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Tools. The computations use PARI/GP [14] for curve data and traces of Frobenius, Arb [1] through `python-flint` for ball arithmetic, and Rust for the floating-point optimiser and the interval prime sum. Claude (Anthropic) was used as a tool in preparing this work. The author takes full responsibility for its content; all results can be reproduced from the code and certificates in the repository.

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Independent researcher (MEGALODON Research), Germany
Email address: `meg.depth@proton.me`